Bing Wu

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EMPLOYMENT

Huazhong University of Science and Technology (HUST) — Wuhan, China
School of Computer Science and Technology
Lecturer — Apr. 2026 - Present

Huazhong University of Science and Technology — Wuhan, China
School of Computer Science and Technology
Data Storage and Application Laboratory
Postdoc — Nov. 2022 - Apr. 2026

EDUCATION

Huazhong University of Science and Technology — Wuhan, China
Ph.D. in Computer Architecture — Sep. 2022
Tutor: *Dan Feng*
Thesis: *Research on ReRAM-based Efficient Technology and Simulation for Processing-in-Memory System*

Huazhong University of Science and Technology — Wuhan, China
B.E. in Computer Science and Technology — Jun. 2016

CONFERENCE PUBLICATIONS

Jinpeng Liu, Shiyi Song, Bing Wu, Huan Cheng, Heng Zhou, Xueliang Wei, Wei Tong, and Dan Feng. An IR drop-robust Mapping Method for Reliable Memristive Accelerators. In *Proceedings of the 2026 Design, Automation and Test in Europe Conference (DATE)*. Apr. 2026.

Taoming Lei, Heng Zhou, Bing Wu, Wei Tong, and Dan Feng. Data Distribution-Aware Analog/Digital Conversion Strategy for Energy-Efficient Memristive In-Situ Accelerators (short paper). In *Proceedings of the 2026 Design, Automation and Test in Europe Conference (DATE)*. Apr. 2026.

Heng Zhou, Bing Wu, Huan Cheng, Jinpeng Liu, Taoming Lei, Dan Feng, and Wei Tong. DRCTL: A Disorder-Resistant Computation Translation Layer Enhancing the Lifetime and Performance of Memristive CIM Architecture. In *Proceedings of the 2024 57th IEEE/ACM International Symposium on Microarchitecture (MICRO)*. Nov. 2024.

Heng Zhou, Bing Wu, Huan Cheng, Wei Zhao, Xueliang Wei, Jinpeng Liu, Dan Feng, and Wei Tong. ODLPIM: A Write-Optimized and Long-Lifetime ReRAM-Based Accelerator for Online Deep Learning. In *Proceedings of the 2023 Design, Automation and Test in Europe Conference (DATE)*. Apr. 2023.

Jinpeng Liu, Wei Tong, Bing Wu, Huan Cheng, and Chengning Wang. ICON: An IR Drop Compensation Method at OU Granularity with Low Overhead for eNVM-based Accelerators. In *Proceedings of the 2023 IEEE 41st International Conference on Computer Design (ICCD)*. Nov. 2023.

Wei Zhao, Wei Tong, Dan Feng, Jingning Liu, Zhangyu Chen, Jie Xu, Bing Wu, Chengning Wang, and Bo Liu. Improving the Energy Efficiency of STT-MRAM based Approximate Cache. In *Proceedings of the 2021 Design, Automation and Test in Europe Conference (DATE)*. Feb. 2021.

Bing Wu, Mengye Peng, Dan Feng, and Wei Tong. DualFS: A Coordinative Flash File System with Flash Block Dual-mode Switching. In *Proceedings of the 2020 IEEE 38th International Conference on Computer Design (ICCD)*, Oct. 2020. **Best Paper in track Computer Systems**.

Wei Zhao, Wei Tong, Dan Feng, Jingning Liu, Jie Xu, Xueliang Wei, Bing Wu, Chengning Wang, Weilin Zhu, and Bo Liu. Oswrite: Improving the Lifetime of MLC STT-Ram with One-step Write. In *Proceedings of the 36th International Conference on Massive Storage Systems and Technology (MSST)*. May 2020.

Bing Wu, Dan Feng, Wei Tong, Jingning Liu, Chengning Wang, Wei Zhao, and Mengye Peng. ReRAM Crossbar-Based Analog Computing Architecture for Naive Bayesian Engine. In *Proceedings of the 2019 IEEE 37th International Conference on Computer Design (ICCD)*. Nov. 2019. **Best Paper Nomination**.

Shuai Li, Wei Tong, Jingning Liu, Bing Wu, and Yazhi Feng. Accelerating Garbage Collection for 3D MLC Flash Memory with SLC Blocks. In *Proceedings of the 2019 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. Nov. 2019.

Bing Wu, Dan Feng, Wei Tong, Jingning Liu, Shuai Li, Mingshun Yang, Chengning Wang, and Yang Zhang. Aliens: A Novel Hybrid Architecture for Resistive Random-Access Memory. In *Proceedings of the 2018 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. Nov. 2018.

Chengning Wang, Dan Feng, Jingning Liu, Wei Tong, Bing Wu, and Yang Zhang. DAWS: Exploiting Crossbar Characteristics for Improving Write Performance of High Density Resistive Memory. In *Proceedings of the 2017 IEEE International Conference on Computer Design (ICCD)*. Nov. 2017.

Yang Zhang, Dan Feng, Jingning Liu, Wei Tong, Bing Wu, and Caihua Fang. A Novel Reram-based Main Memory Structure for Optimizing Access Latency and Reliability. In *Proceedings of the 2017 54th Design Automation Conference (DAC)*. Jun. 2017.

JOURNAL PUBLICATIONS

Yibo Liu, Bing Wu, Shiyi Song, Xueliang Wei, Huan Cheng, Heng Zhou, Wei Tong, and Dan Feng. pTree: Building Efficient B+-Tree on Non-volatile Memory with Processing-in-memory. In *IEEE Transactions on Computers (TC)*, early access.

Jinpeng Liu, Wei Tong, Bing Wu, Huan Cheng, Heng Zhou, Xueliang Wei, and Dan Feng. ICON-NIV: A Generalized Method for Mitigating the Impacts of IR Drop and Nonlinear I-V Effect in eNVM-based Accelerators. In *ACM Transactions on Architecture and Code Optimization (TACO)*, 2026.

Bing Wu, Yibo Liu, Jinpeng Liu, Huan Cheng, Xueliang Wei, Wei Tong, and Dan Feng. FADESIM: Enable Fast and Accurate Design Exploration for Memristive Accelerators Considering Non-idealities. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025.

Xueliang Wei, Dan Feng, Wei Tong, Bing Wu, and Xu Jiang. COVER: Alleviating Crash-Consistency Error Amplification in Secure Persistent Memory Systems. In *ACM Transactions on Architecture and Code Optimization (TACO)*, 2025.

Wei Zhao, Dan Feng, Wei Tong, Xueliang Wei, and Bing Wu. CMD: A Cache-Assisted GPU Memory Deduplication Architecture. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025.

Xueliang Wei, Dan Feng, Wei Tong, Bing Wu, and Xu Jiang. SEED: Speculative Security Metadata Updates for Low-Latency Secure Memory. In *ACM Transactions on Architecture and Code Optimization (TACO)*, 2025.

Wei Zhao, Dan Feng, Wei Tong, Jingning Liu, Zhangyu Chen, Bing Wu, and Chengning Wang. APPcache+: An STT-MRAM-Based Approximate Cache System with Low Power and Long Lifetime. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2023.

Wei Zhao, Jie Xu, Xueliang Wei, Bing Wu, Chengning Wang, Weilin Zhu, Wei Tong, Dan Feng, and Jingning Liu. A Low-latency and High-endurance MLC STT-MRAM-based Cache System. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2022.

Chengning Wang, Dan Feng, Wei Tong, Jingning Liu, Bing Wu, and Yiran Chen. Space-Time-Efficient Modeling of Large-Scale 3-D Cross-Point Memory Arrays by Operation Adaption and Network Compaction. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2021.

Chengning Wang, Dan Feng, Wei Tong, Jingning Liu, Bing Wu, Wei Zhao, Yang Zhang, and Yiran Chen. Improving Write Performance on Cross-point RRAM Arrays by Leveraging Multidimensional Non-uniformity of Cell Effective Voltage. In *IEEE Transactions on Computers (TC)*, 2020.

Chengning Wang, Dan Feng, Wei Tong, Yu Hua, Jingning Liu, Bing Wu, Wei Zhao, Linghao Song, Yang Zhang, Jie Xu, Xueliang Wei, and Yiran Chen. Improving Multilevel Writes on Vertical 3-D Cross-point Resistive Memory. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2020.

Bing Wu, Dan Feng, Wei Tong, Jingning Liu, Chengning Wang, Wei Zhao, and Yang Zhang. A Low Power Reconfigurable Memory Architecture for Complementary Resistive Switches. In *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2019.

Chengning Wang, Dan Feng, Wei Tong, Jingning Liu, Zheng Li, Jiayi Chang, Yang Zhang, Bing Wu, Jie Xu, Wei Zhao, Yilin Li, and Ruoxi Ren. Cross-point Resistive Memory: Nonideal Properties and Solutions. In *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2019.

Chengning Wang, Dan Feng, Wei Tong, Jingning Liu, Bing Wu, Wei Zhao, and Yang Zhang. Design and Analysis of Address-adaptive Read Reference Settings for Multilevel Cell Cross-point Memory Arrays. In *IEEE Transactions on Electron Devices (TED)*, 2019.

Yang Zhang, Dan Feng, Wei Tong, Yu Hua, Jingning Liu, Zhipeng Tan, Chengning Wang, Bing Wu, Zheng Li, and Gaoxiang Xu. CACF: A Novel Circuit Architecture Co-optimization Framework for Improving Performance, Reliability and Energy of ReRAM-based Main Memory System. In *ACM Transactions on Architecture and Code Optimization (TACO)*, 2018.

AWARDS

Silver Award in the 3rd China Postdoctoral Innovation & Entrepreneurship Competition 2025

ICCD Best Paper in Track Computer Systems 2020

ICCD Best Paper Nomination 2019

National Scholarship for Doctoral Students 2019

The First Prize of National Olympiad in Informatics in Provinces (NOIP) 2010 & 2011

FUNDS AND PROJECTS

Research on Multi-level Co-optimization for Improving Memristive Processing-in-Memory System Reliability (No. 62302179). Source: National Natural Science Foundation of China (NSFC), 01/2024 - 12/2026. Role: PI.

SERVICE

Journal Reviewer

IEEE Transactions on Computers (TC) 2026

IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS) 2022

TEACHING

Design and Analysis of Algorithms (CST2261), and its Practice (CST0032) HUST

Instructor Fall 2026

Fundamentals of Software Technique (SFT0015) HUST

Instructor Fall 2026